

Lid-Monitoring SENSOR MODULE

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If you need a lid-monitoring sensor for an application, this is the circuit you have been looking for. One application for such sensors would be in the security market. For example, construction companies need more ways to combat tools theft. Construction toolboxes typically remain on a construction site when the workers leave, making these easy targets for theft. Lid-monitoring sensors can be used to detect

when a toolbox is being opened or tampered with, among other things, which can be used in conjunction with a microcontroller (MCU) based monitoring device.

Here is a simple lid-monitoring sensor module circuit realised using an inexpensive mercury-free tilt switch. The standalone roller-ball-type tilt sensor senses tilt angle/movement, and its overall response is very fast.

of two conductive elements (poles) and a conductive free mass (rolling ball), encapsulated in the same case. When the tilt sensor is oriented so that that end is downwards, the mass rolls onto the poles and shorts these, acting as a switch throw. Working principle of SW-520D tilt sensor is shown in Fig. 2.

Rest of the circuit is based on a 'misused' LM393 (IC1) configured as a simple RC op-amp comparator timer. In idle state, voltage is applied across the R1-C1 combination, producing a typical RC charging signal

Table I: Tilt Sensor and Circuit Output

Tilt sensor/input socket (CON1)	Pin 2 of output socket (CON2)
Off/open	High
On/short	Low

Circuit and working

A look at Fig. 1 will convince you that the circuit diagram of the lid-monitoring sensor module is pleasantly simple. Admittedly, a roller-ball-type tilt sensor (SW-520D) is used here. Such a device consists

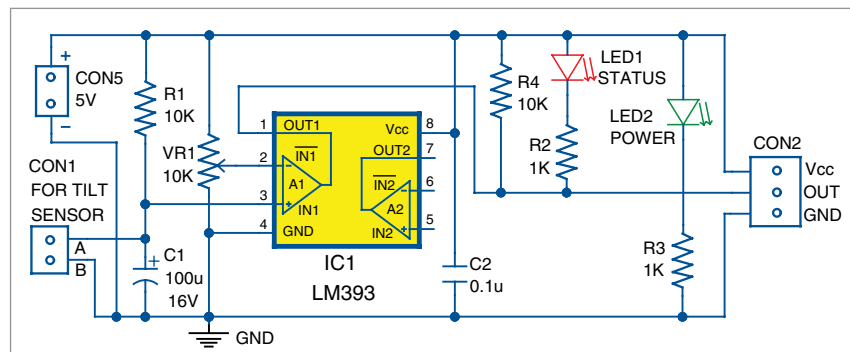


Fig. 1: Circuit of the lid-monitoring sensor module

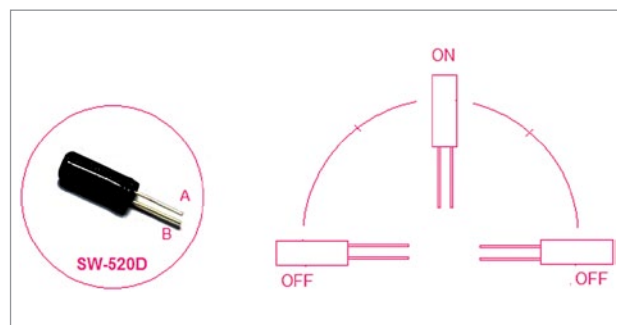
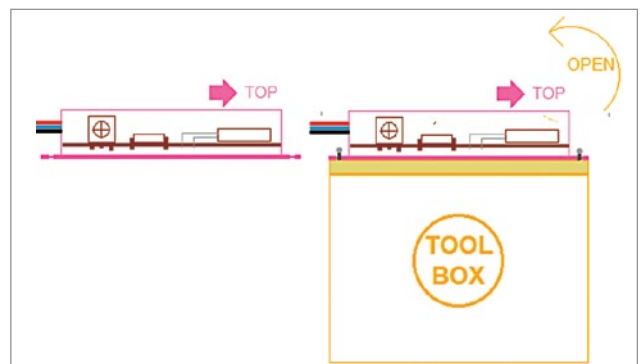


Fig. 2: Tilt sensor and its working



Figs 3 and 4: Proposed enclosure and its placement